

Project #1

Plastic Scanner Initial Checks

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1 Introduction

This report is the first in a series of five, about the use of a Nordic Semiconductor nRF52 Development Kit (nRF52 DK) along with a Printed Circuit Board (PCB) for scanning plastics. The PCB is designed for use with an Arduino Uno rev. 3 [1]. This report will investigate whether the PCB is built according to given specifications, as well as whether the nRF52 DK and PCB are compatible with one another.

1.1 Verification of component placement

This section will look at whether or not components are correctly placed on the PCB. In summary:

- D1-D5, D12 are all present and correctly placed on the PCB.
- D10, D11 are missing as mentioned in BOM.
- U1-U4 are all present and correctly placed on the PCB.
- No component is misplaced or incorrectly soldered.

This will be explained further, starting with the placement of diodes.

1.1.1 Placement of diodes

Figure 1 shows how the LEDs should be mounted on the PCB, indicating what is the correct orientation when mounted. Note that all LEDs are connected to the same node. Figure 2 shows how the LEDs are mounted. Note that the markings, as indicated by the blue arrows, are on the cathode for D4 and D5, and on the anode for D6-D9. This is shown in fig. 3, which contains excerpts from the user manuals for these LEDs. It also shows that D10 and D11 is missing, which is in accordance with the bill of materials.

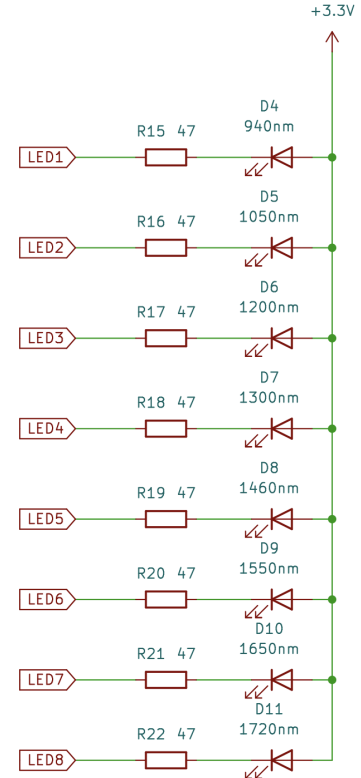


Figure 1: LED schematic [1].

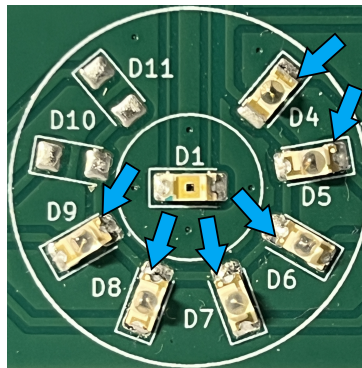


Figure 2: LEDs on the board.

Peak Emission Wavelength: 1200nm



Description

- Size 1206: 3.2 (L) x 1.6 (W) x 1.9 (H) mm
- Circuit substrate: glass laminated epoxy
- Devices are RoHS and REACH conform
- Lead free solderable, soldering pads: gold plated
- Marking at anode
- High radiation intensity types

(a) D6

Peak Emission Wavelength: 1300nm



Description

- Size 1206: 3.2 (L) x 1.6 (W) x 1.95(H) mm
- Circuit substrate: glass laminated epoxy
- Devices are RoHS conform
- Lead free solderable, soldering pads: gold plated
- Marking at anode
- High radiation intensity types

(b) D7

Peak Emission Wavelength: 1460nm

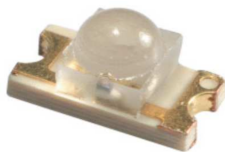


Description

- Size 1206: 3.2 (L) x 1.6 (W) x 1.95(H) mm
- Circuit substrate: glass laminated epoxy
- Devices are RoHS conform
- Lead free solderable, soldering pads: gold plated
- Marking at anode
- High radiation intensity types

(c) D8

Peak Emission Wavelength: 1550nm



Description

- Size 1206: 3.2 (L) x 1.6 (W) x 1.2 (H) mm
- Circuit substrate: glass laminated epoxy
- Devices are RoHS conform
- Lead free solderable, soldering pads: gold plated
- Marking at anode
- High radiation intensity

(d) D9

Figure 3: LED documentation stating markings are at anode [2, 3, 4, 5].

1.1.2 Placement of chips

The chips are also correctly mounted, as indicated by the orientation marks. Figure 4 shows this.

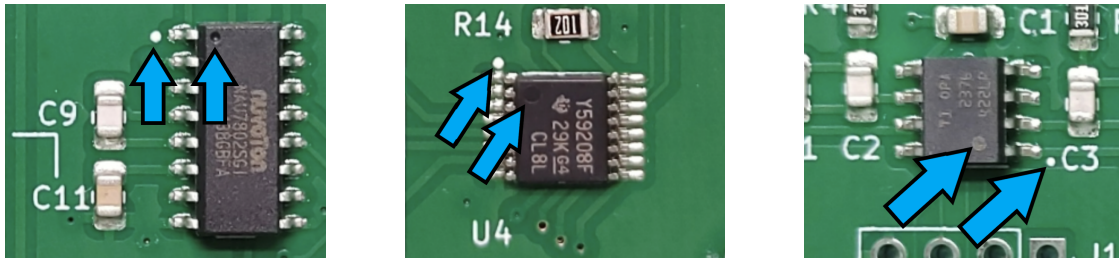


Figure 4: Chips on the PCB.

1.2 Compatability analysis

As shown in fig. 5, the nRF52 DK should be compatible with 3rd party Arduino shields. The plastic scanner PCB is such a shield, and it follows that the nRF52 DK can replace the Arduino Uno Rev. 3 in the plastic scanner. The compatability will however be investigated a bit further.

KEY FEATURES

- Affordable, rapid prototyping and development solution for the nRF52810 and nRF52832 SoCs
- Supports Bluetooth LE, Bluetooth mesh, ANT and 2.4 GHz proprietary development
- Bluetooth 5 feature support: 2 Mbps, Advertising Extensions and Channel Selection Algorithm #2 (CSA #2)
- Arm® Cortex™ M4 with floating point support
- DSP instruction set
- Arduino Uno Rev. 3 compatible for use with 3rd party shields
- All I/Os and interfaces are available via connectors

Figure 5: Key features of nRF52 DK [6].

1.2.1 Absolute maximum ratings

	Min.	Max.	Unit
Supply voltages			
VDD	-0.3	+3.9	V
VSS		0	V
I/O pin voltage			
V _{I/O} , VDD ≤ 3.6 V	-0.3	VDD + 0.3 V	V
V _{I/O} , VDD > 3.6 V	-0.3	3.9 V	V

Figure 6: Absolute maximum ratings for the nRF52832 [7].

Figure 6 shows the absolute maximum ratings for the nRF52 DK IO pins to be 3.9V. The plastic scanner is built for 5V input, but as fig. 7 shows, the 5V input is adjusted down to 3.3V, which is within the absolute maximum ratings.

1.2.2 Logic voltage

Figure 8 shows a warning (blue text) about the logic voltage for the Arduino being at 5V, which further indicates that a lower voltage is preferable for the plastic scanner. As shown in fig. 9, the MCU_RESET and TRIGGER are both pulled up to the 5V raw flag when buttons are inactive, which is at 5V when the development board is connected to

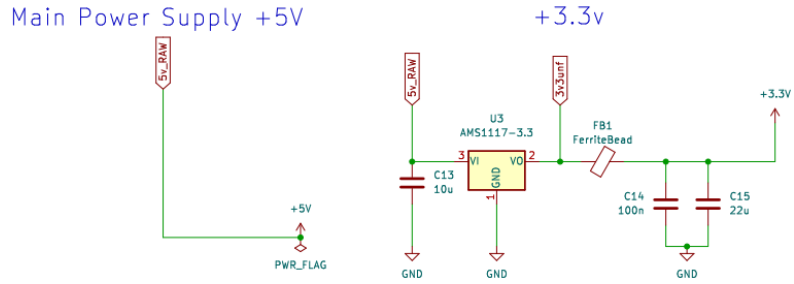


Figure 7: Plastic scanner power supply [1].

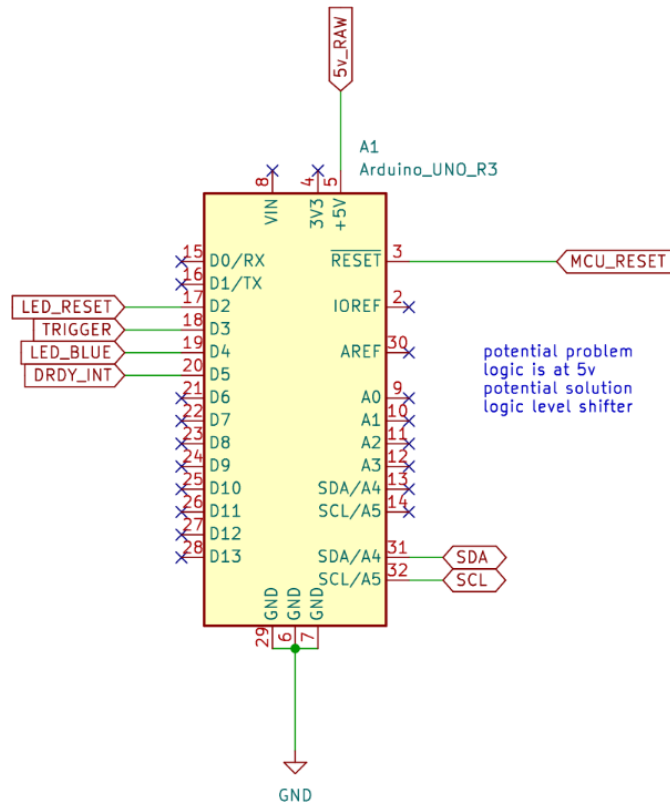


Figure 8: Arduino schematic [1].

the USB port. This is a potential issue as the voltage exceeds the nRF52 DK I/O pin max voltage of 3.9V.

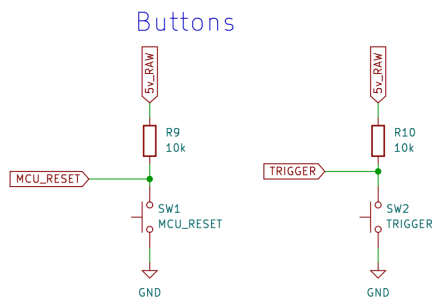


Figure 9: Arduino schematic [1].

2 Discussion

2.1 Markings on LEDs

LEDs are commonly marked on the cathode, however in the case of D6-D9 the markings are on the anode. The PCB was originally soldered based on the assumption that markings would be on the cathode, and so they had to be re-soldered. This is shown in fig. 10.

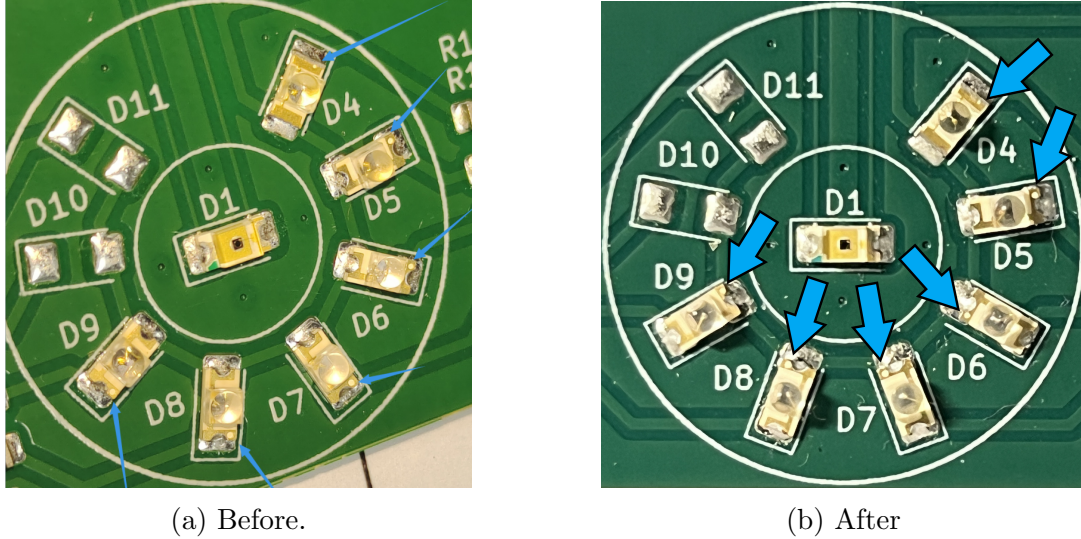


Figure 10: LEDs before and after resoldering.

2.2 Logic at 5V

As stated above, MCU_RESET and TRIGGER are pulled up to the 5V raw flag, which is an issue as the voltage exceeds the nRF52832 I/O pin max voltage of 3.9V. A voltage divider can be used to reduce the voltage to 3.3V, but it would require some changes to the PCB design. Another option, and the one that will be practiced, is to simply ignore the issue and use it as it is.

3 Conclusion

The plastic scanner was not built correctly according to spec initially, because four of the LEDs were soldered the wrong way around. However after re-soldering, it is built correctly. The nRF52 DK and the plastic scanner are not compatible on paper, due to MCU_RESET and TRIGGER exceeding the maximum I/O voltage for the nRF52832. This issue will be overlooked, and the plastic scanner will be used with the nRF52 DK.

References

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